Hanfei Ding

+86 188-1451-5758 | dinghanfei1127@163.com | dinghanfei.github.io | github.com/dinghanfei

Education

Harbin Engineering University (Project 211) **Sep. 2021 – Jun. 2025**

Bachelor of Engineering in Computer Science and Technology *Harbin, China*

- **GPA: 90.65/100**
  Achieved scores of **90+** in **30 required courses**, including **92+** in all **mathematics-related courses**.
- **TOEFL iBT: 4.5/6.0**

Relevant Coursework

- Probability and Statistics (**100**)
- Mathematical Analysis (**98**)
- Linear Algebra (**97**)
- Data Structures (**95**)
- Algorithm Design and Analysis (**93**)
- Operating Systems (**91**)
- Computer Organization (**96**)
- Artificial Intelligence (**95**)
- Discrete Mathematics (**92**)

Research Experience

Parameter-Efficient Adaptation of SAM for Brain Tumor Segmentation **Dec. 2024 – Jan. 2025**

Undergraduate Research Assistant (Team Leader), Deep Intelligence Laboratory, Harbin Engineering University *China*

- Led the adaptation and evaluation of SAM-based models for multimodal brain MRI segmentation on the BraTS 2020 dataset, investigating domain-specific adaptation under limited labeled medical data.
- Designed a modality-aware preprocessing pipeline by converting 3D MRI volumes into normalized pseudo-RGB slices from FLAIR, T1ce, and T2 modalities, generating WT/TC/ET masks and patient-level splits to prevent data leakage.
- Implemented parameter-efficient fine-tuning by freezing the image encoder and prompt encoder while optimizing only the mask decoder with stochastic box prompts and BCE-Dice loss, reducing trainable parameters and overfitting risk.
- Validated the effectiveness of SAM-based adaptation on BraTS 2020, achieving an average Dice score of 0.7984 with MedSAM and developing an interactive web-based prototype for prompt-guided tumor segmentation.

Masked Face Recognition with Machine Learning and OpenCV **May 2023 – Jul. 2023**

Team Leader, Visual Computing Summer Workshop, ***National University of Singapore*** *Singapore*

- Developed an OpenCV-based face recognition system for masked and unmasked faces, addressing recognition degradation under facial occlusion.
- Built a masked-face dataset using dlib facial landmarks, generated Gaussian-blurred and mean-filtered variants, and extracted SIFT descriptors for robust facial representation.
- Configured five decision branches using SVM, KNN, and Random Forest with different preprocessing strategies and combined their outputs through a rule-based decision scheme, correctly rejecting 100% of unknown users, accepting 86% of registered users, and achieving 85% overall accuracy.

Projects

Multimodal Fault Diagnosis for Microservice Systems | *PyTorch, GAT, GGNN* **Mar. 2025 – Jun. 2025**

- Independently designed and implemented a three-stage multimodal fault diagnosis tool as an undergraduate thesis project, integrating logs, traces, and metrics for joint anomaly detection and root cause localization.
- Encoded heterogeneous observability signals with modality-specific encoders and GLU-based adaptive fusion to generate service-level node embeddings.
- Constructed service invocation graphs and learned topology-aware representations with GAT and GGNN to capture dynamic service dependencies and long-range fault propagation.
- Developed a parameter-shared joint classifier for cascaded anomaly detection and root cause localization, achieving the best overall performance among multiple baselines across five evaluation metrics on the Eadro dataset and receiving the **Outstanding Undergraduate Thesis (Design) Award**.

Evidence-Grounded Deep Research Agent | *LangGraph, Agentic RAG, Hybrid Retrieval, FastAPI* **May 2025**

- Built an evidence-grounded research agent that decomposes open-ended research questions, retrieves web evidence, identifies information gaps, and generates structured reports with traceable citations.
- Built a LangGraph workflow with structured state, execution budgets, and fair scheduling, raising research-task coverage from 32% to 94%; added checkpoint recovery, citation validation, role-based handoffs, and an async FastAPI service.
- Designed language-aware hybrid retrieval by fusing dense and BM25 results via RRF, followed by multilingual reranking, achieving 100% Recall@5, 95% MRR, and 96.31% NDCG@5 on a 30-query offline benchmark.

Awards & Honors

Competitions

- **Mathematical Modeling League of Three Northeastern Provinces** (Top 18.52%) **Jul. 2024**
  Regional 1st Prize; **Team Leader / Primary Contributor**
- **Lanqiao Cup Software Competition** (Top 20%) **May 2024**
  Provincial 2nd Prize
- **National English Competition for College Students, Category C** (Top 8.6%) **Oct. 2022**
  National 3rd Prize
- **China Undergraduate Mathematical Contest in Modeling** (Top 0.6% among 49,000+ entries) **Sep. 2022**
  Provincial 1st Prize; **Team Leader / Primary Contributor**

Scholarships & Honors

- **Outstanding Undergraduate Thesis (Design), Class of 2025** **Jun. 2025**
  One of 11 recipients across the School of Computer Science and Technology, School of Software Engineering, and School of Information Security.
- **Outstanding Student Scholarship, Harbin Engineering University** **Multiple Years**
  School-level scholarship; 5-time recipient; awarded to the top 8–15% of students.
- **Merit Student, Harbin Engineering University** **2021 – 2022**
  University-level honor; awarded to the top 6% of students.

Extracurricular Activities

Underwater Robotics Club | *Mechanical Team Member* **Nov. 2021 – Sep. 2022**

- Collaborated with a multidisciplinary robotics team and contributed to the structural design of underwater robotic systems, developing hands-on engineering experience and teamwork skills through practical projects.

Freshman Debate Competition | *Debate Team Member* **Oct. 2021**

- Represented the department in interdepartmental debate competitions and participated in multiple internal training sessions, receiving the **Best Debater** award for outstanding reasoning and communication performance.

Campus Volunteer Service | *Campaign Volunteer* **Sep. 2021**

- Participated in university volunteer campaigns promoting low-carbon practices and public health awareness, contributing to community engagement initiatives and social responsibility activities.

Skills & Interests

Programming Languages: C, C++, Python, Java
Frameworks & Tools: PyTorch, OpenCV, scikit-learn, LangGraph, FastAPI, Chroma, Git, Docker, Linux
Interests: swimming, badminton, fitness training, LEGO model building